

The RateBeer Case: Debrief and Part 2

MKT 566 · Fall 2026

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Before We Start

Housekeeping

- **Today** is the deadline for being in a group: add yours to the [sign-up sheet](#)
- The **RateBeer report** ([ratebeer-report.html](#) + [ai-log.md](#)) is due **Sunday, Sept. 20, 11:59 pm** on Brightspace
- Thursday: **regression** (OLS), the tool behind every fitted line you drew last week

Today's plan

1. **Debrief of Tasks 1–6** (about 25 minutes): chart, “what do you see?”, then the reading. Compare with your own answers as we go
2. **Tasks 7–9** on your laptop: the brewery table, time, and the recommendation
3. Last 10 minutes: the **report checklist**

Keep your `ratebeer-case.R` open. If a chart of yours looks different from mine, that is worth a question, not a worry.

Tasks 1–3: Look, Find the Problems, Clean

Task 1: what is a row?

```
1 head(raw[, .(beer_brewerId, beer_style, beer_ABV, review_overall, review_time)], 4)
```

	beer_brewerId	beer_style	beer_ABV	review_overall	review_time
	<int>	<char>	<char>	<char>	<int>
1:	8481	Klsch	5.4	8/20	1157587200
2:	3228	Sour Ale/Wild Ale	–	17/20	1266019200
3:	3228	Traditional Ale	6.7	16/20	1237420800
4:	3228	Traditional Ale	6.7	14/20	1229040000

```
1 nrow(raw)
```

```
[1] 292680
```

```
1 uniqueN(raw$beer_beerId); uniqueN(raw$beer_brewerId); uniqueN(raw$review_profileName)
```

```
[1] 50826
```

```
[1] 5841
```

```
[1] 13030
```

292,680 rows. What does one row represent, and why does it matter?

Task 1: reading it

- One row is one **review**: one person rating one beer once
- Behind the 292,680 rows: **50,826** beers, **5,841** breweries, **13,030** reviewers
- So “reviews by style” counts a beer **once per review**. A style with a few famous beers can look bigger than a style with many quiet ones

The **unit of observation** is the first thing to settle in any dataset. Every count and every average depends on it. We change it on purpose in Task 7.

Task 2: find the problems

```
1 for (col in c("beer_ABV", "review_overall", "review_time")) {  
2   cat("\n==", col, "(", class(raw[[col]]), ")\n")  
3   print(head(sort(table(raw[[col]]), decreasing = TRUE), 5))  
4 }
```

== beer_ABV (character)

	5	-	5.5	6	8
	25193	14140	11566	11129	10630

== review_overall (character)

	14/20	15/20	13/20	16/20	12/20
	43254	41840	35988	33424	30224

== review_time (integer)

	1188604800	1227139200	1296950400	1322265600	1299369600
	540	256	213	208	205

Three columns, three problems. What are they?

Task 2: reading it

- **Ratings are text:** "14/20" is five characters, not a number. R cannot average text
- **beer_ABV has "-"** in 14,140 rows. It means **missing**, not zero. Real alcohol-free beers are in the data at 0.0–0.5, so 14,140 zeros would be wrong and would pull the average down
- **review_time is a Unix timestamp:** seconds since January 1, 1970. 1157587200 is September 7, 2006
- And in **beer_style**: sake (12 kinds), cider, perry, and mead are not beer; **Klsch** and **(IPA)** are encoding damage from the scraper

None of this is visible in a chart. You find it by **looking at the values** before you plot anything. That is what Tasks 1–2 are for.

Task 3: after cleaning

```
1 n_raw - nrow(beer); nrow(beer)
```

```
[1] 3693
```

```
[1] 288987
```

```
1 summary(beer[, .(review_overall, review_taste, review_appearance, beer_ABV)])
```

review_overall	review_taste	review_appearance	beer_ABV
Min. : 1.0	Min. : 1.000	Min. : 1.000	Min. : 0.010
1st Qu.: 12.0	1st Qu.: 6.000	1st Qu.: 3.000	1st Qu.: 5.000
Median : 14.0	Median : 7.000	Median : 3.000	Median : 5.900
Mean : 13.2	Mean : 6.451	Mean : 3.436	Mean : 6.616
3rd Qu.: 15.0	3rd Qu.: 8.000	3rd Qu.: 4.000	3rd Qu.: 8.000
Max. : 20.0	Max. : 10.000	Max. : 5.000	Max. : 57.700
			NAs : 14019

```
1 range(beer$review_date)
```

```
[1] "2000-04-26" "2012-01-13"
```

What is the scale of each rating? How many ABV values are missing?

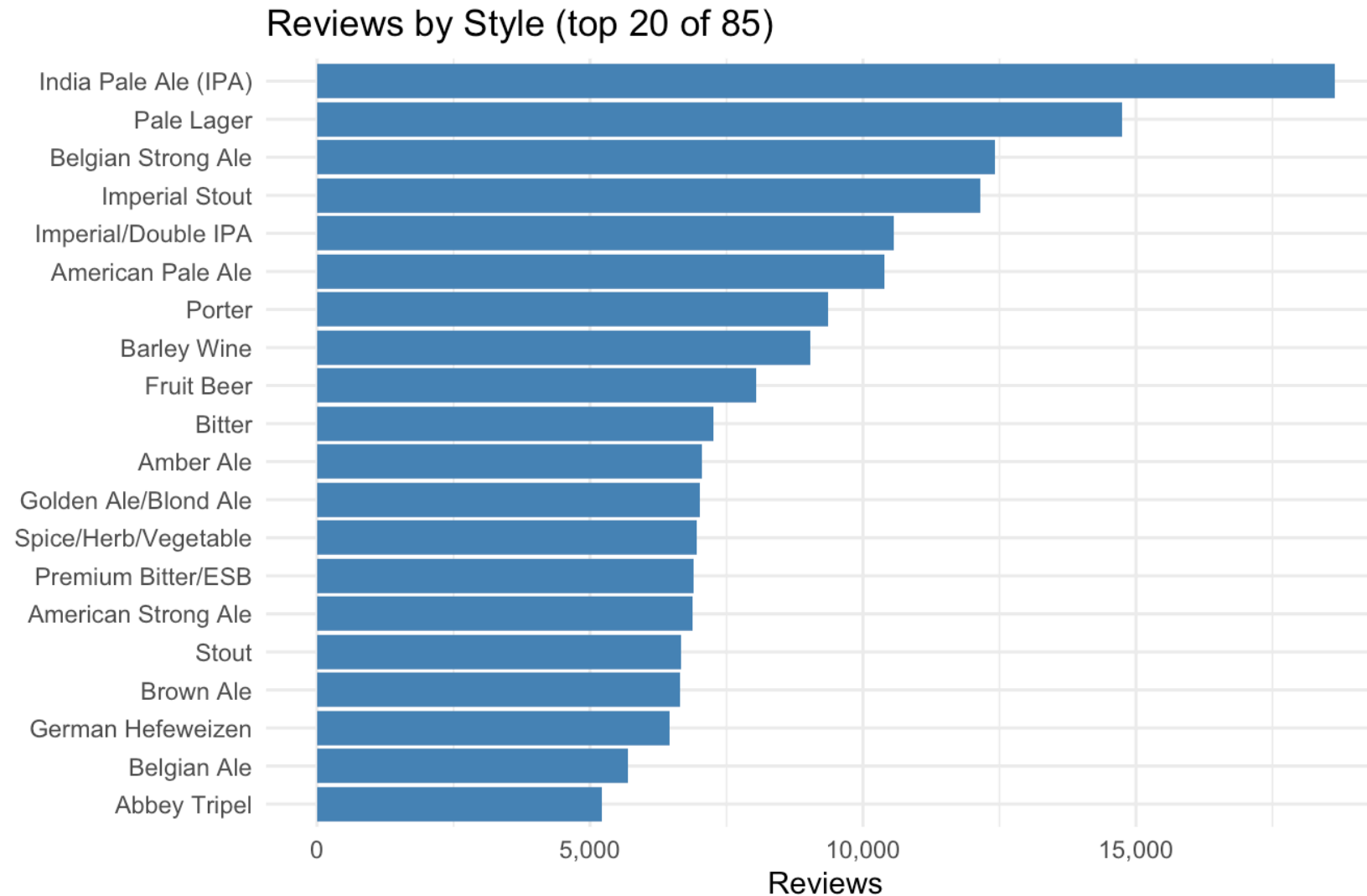
Task 3: reading it

- Non-beers removed: **3,693** rows, leaving **288,987**
- Scales differ: overall **1–20**, taste and aroma **1–10**, appearance and palate **1–5**. A taste of 7 and an overall of 14 are both 70% of their scale; you cannot compare the raw numbers
- **14,019** reviews have no ABV: keep them for everything about ratings, drop them only in ABV charts (`na.rm = TRUE`)
- Reviews run from April 2000 to **January 13, 2012**. Remember that date for Task 8

Did you check the AI's work? Mean overall should be **13.2**. If yours was 0.7, the assistant kept the `/20` part. The assistant cannot see your output.

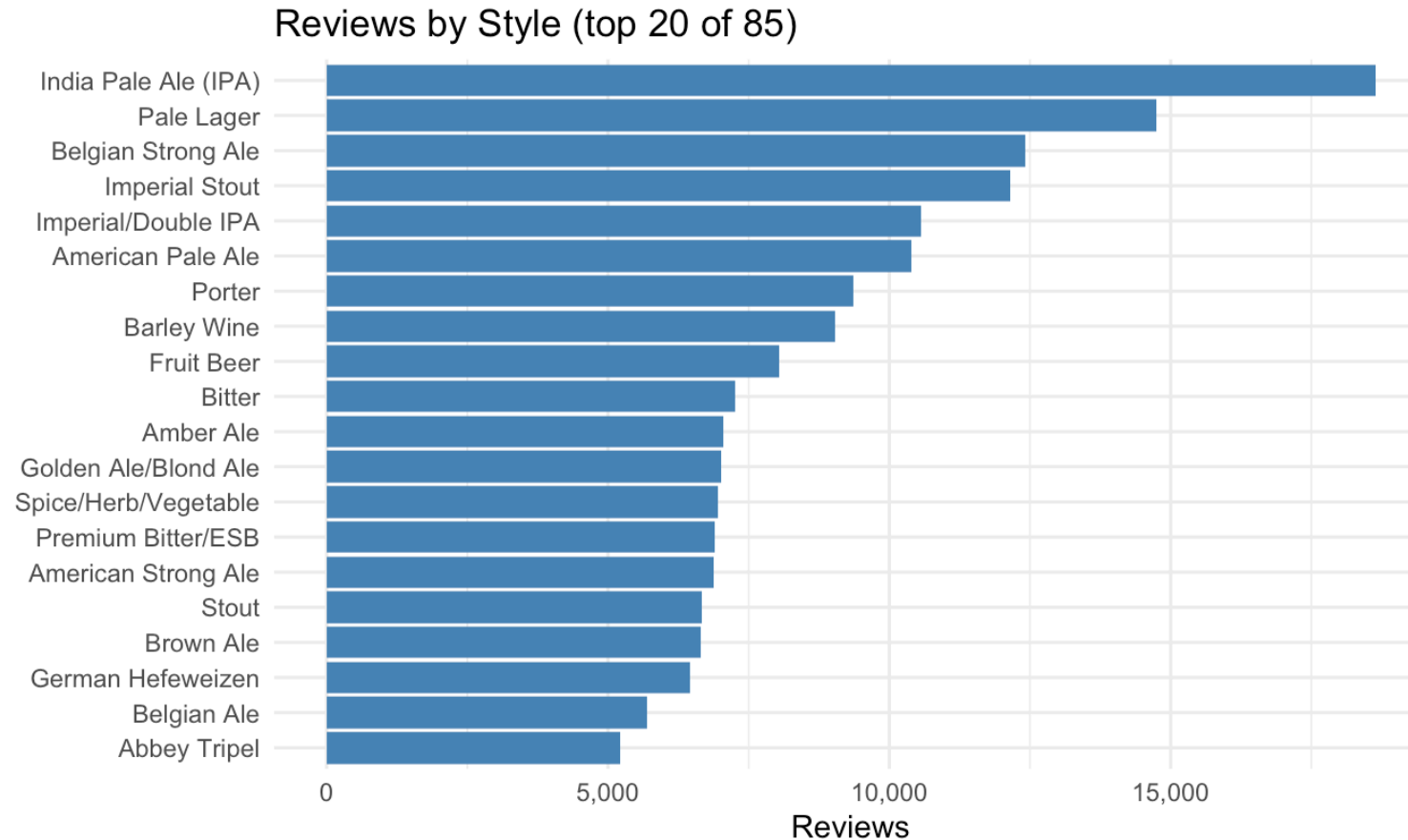
Task 4: Variation Warm-up

Reviews by style



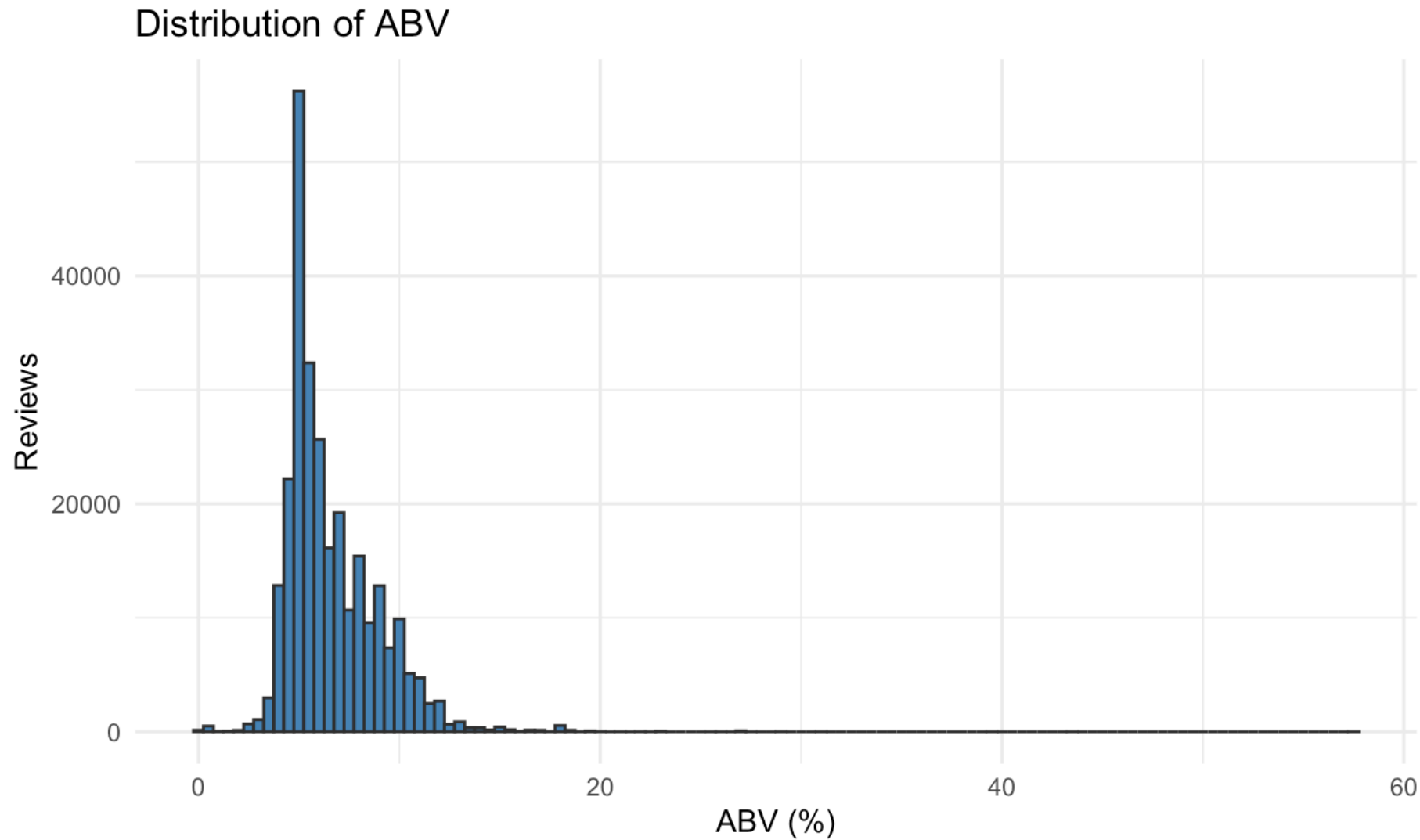
Which styles dominate? And what does “most reviewed” actually measure?

Reviews by style: reading it



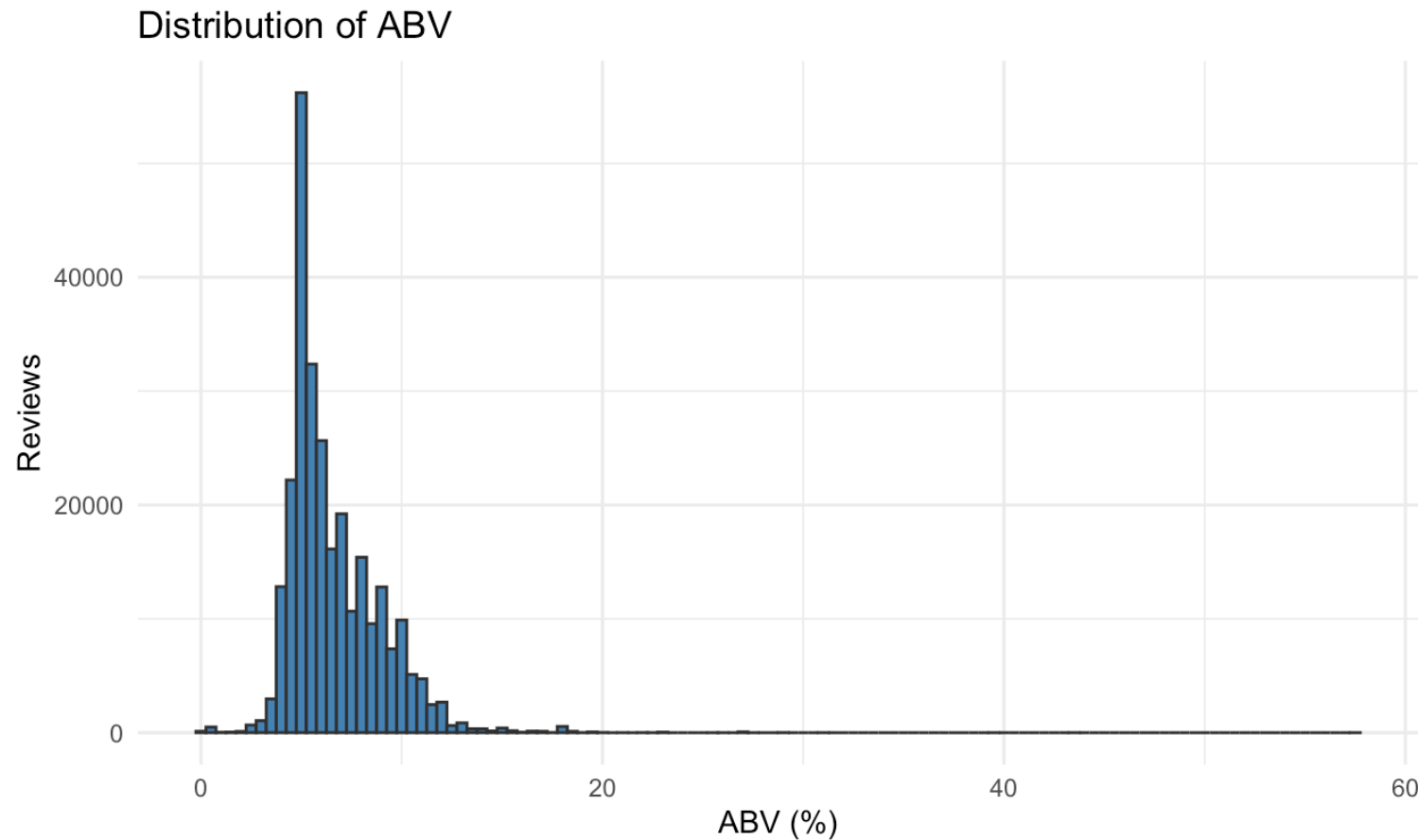
- **IPA** leads (18,635), then Pale Lager, Belgian Strong Ale, Imperial Stout
- Pale Lager is the most **consumed** beer in the world, and it is second here only because everyone has tried one. This is what **enthusiasts** write about, not what the market buys. Keep that in mind for Task 9

The distribution of ABV



What shape? And what is hiding at the far right?

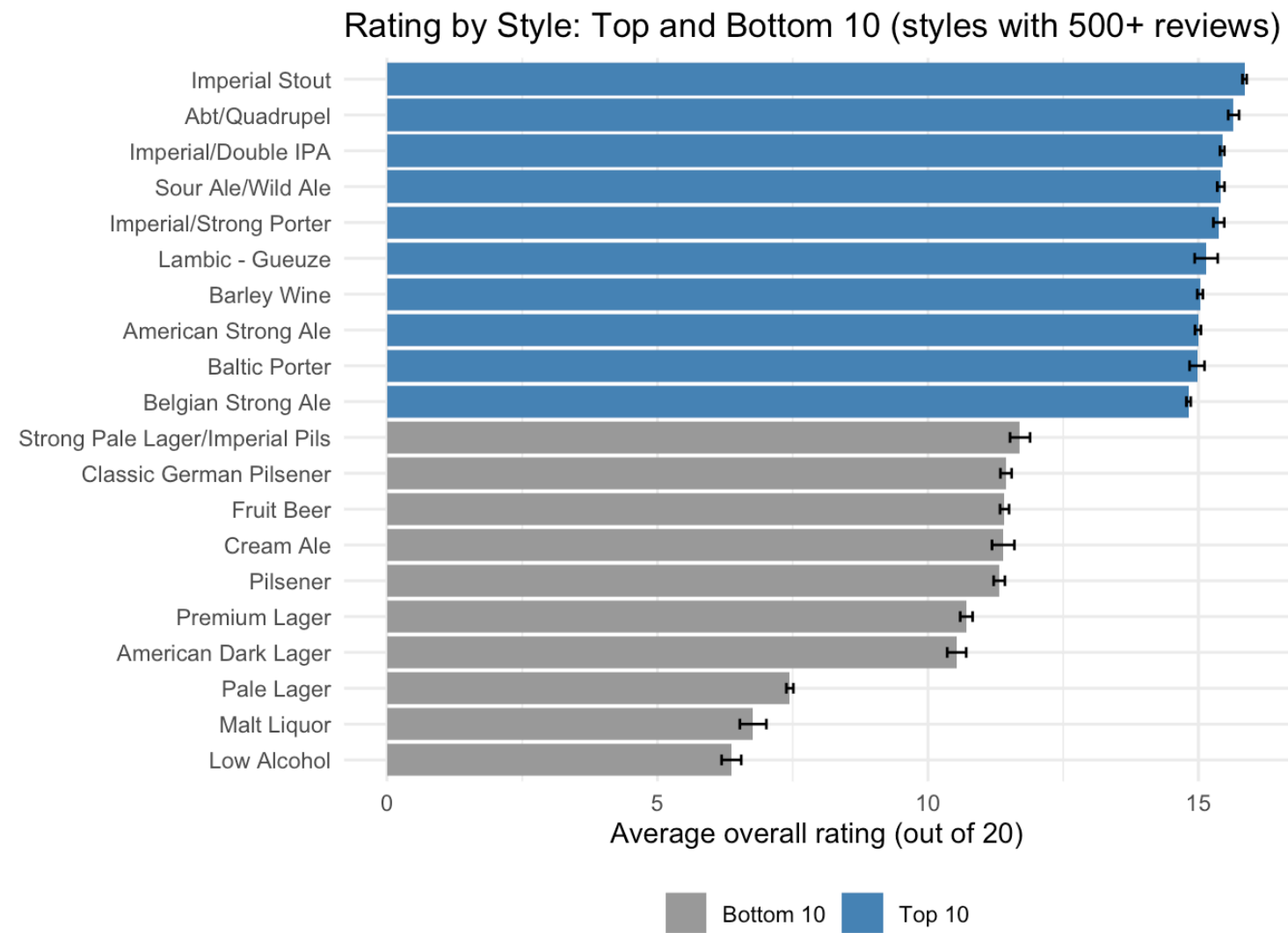
The distribution of ABV: reading it



- **Right-skewed**: most reviews are of beers between 4% and 8%, with a long tail past 10%
- The maximum is **57.7%**: Schorschbräu Schorschbock, a real freeze-distilled Eisbock, not a typo. An outlier is not automatically an error. **Check before you drop**

Task 5: Rating by Style

Average rating by style



Top three, bottom three? Why are the error bars so small? (Your chart has all 50 styles; this shows the two ends.)

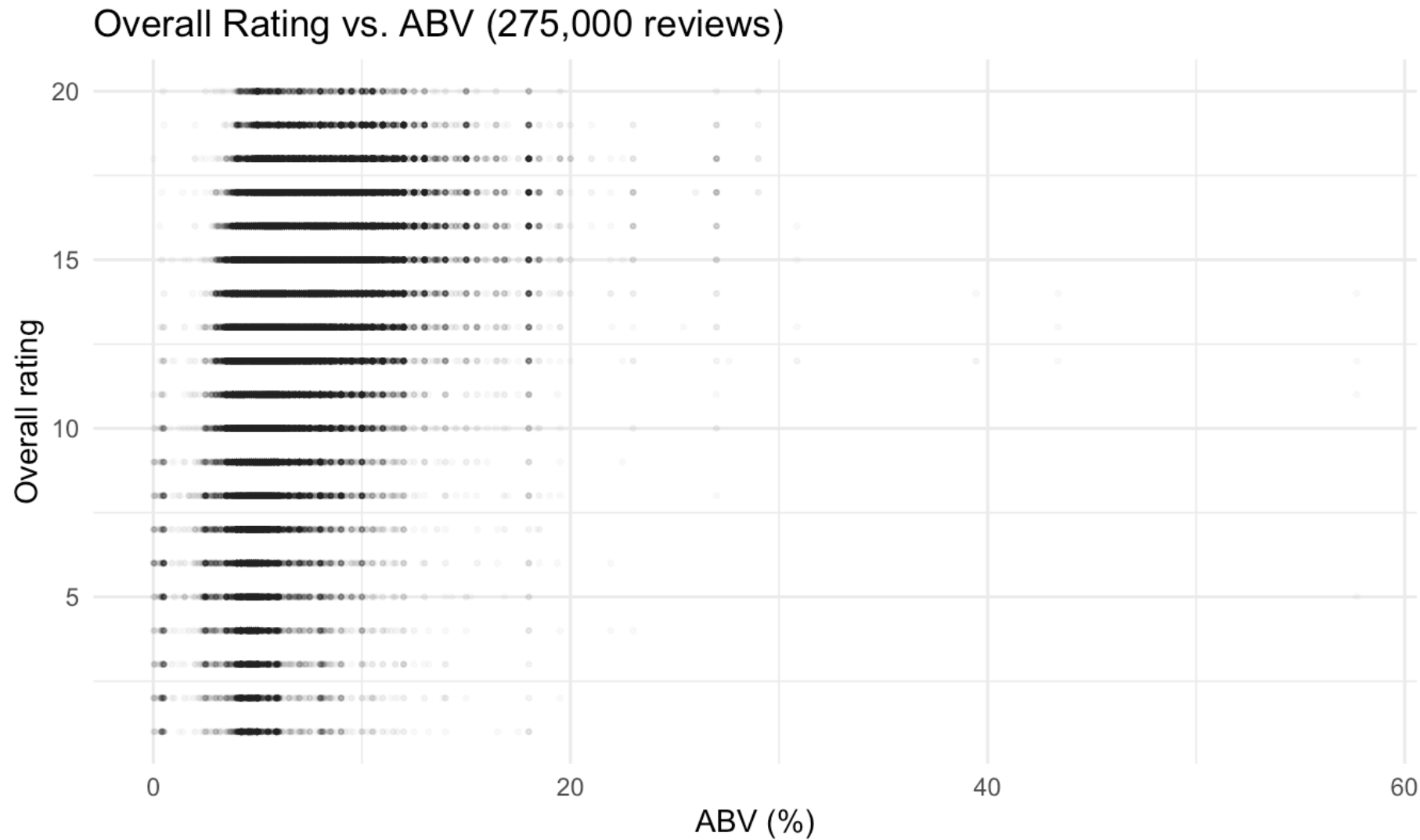
Average rating by style: reading it

	beer_style	avg	n	abv
	<char>	<num>	<int>	<num>
1:	Imperial Stout	15.85	12157	10.2
2:	Abt/Quadrupel	15.65	1966	10.4
3:	Imperial/Double IPA	15.44	10559	9.1
4:	Pale Lager	7.45	14754	4.7
5:	Malt Liquor	6.77	1489	7.2
6:	Low Alcohol	6.37	1264	1.2

- **Imperial Stout, Quadrupel, Double IPA** at the top (15.4–15.9); **Pale Lager, Malt Liquor, Low Alcohol** at the bottom (6.4–7.4). Less than half
- Thousands of reviews per style, so the error bars are tiny: these gaps are not noise. The 500-review rule keeps a style with 12 lucky reviews off the top
- Look at the ABV column: the top styles are all **strong** (9–11%), the bottom ones light. Style and alcohol are tangled together. Next task

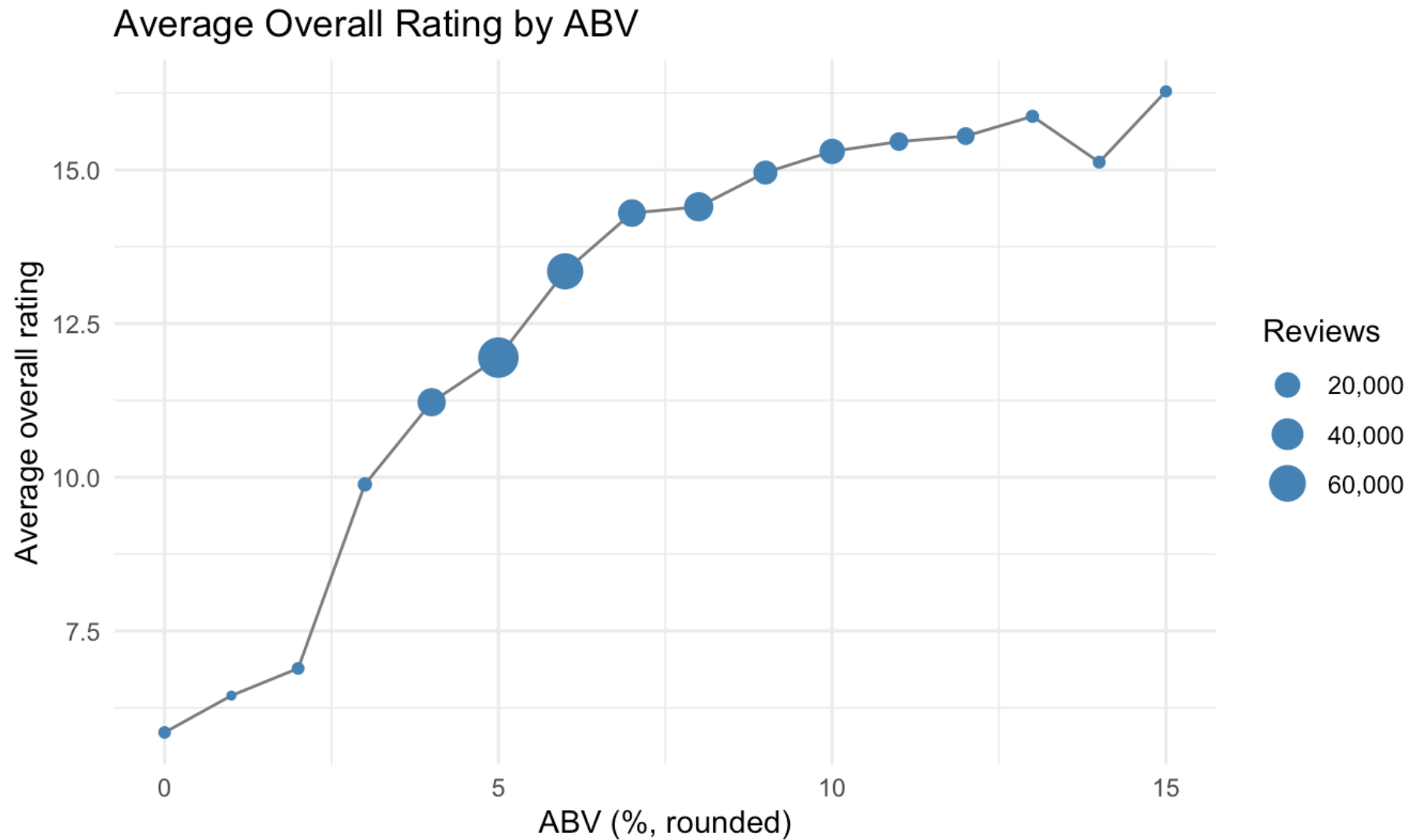
Task 6: Rating vs. Alcohol

First attempt: a scatter plot



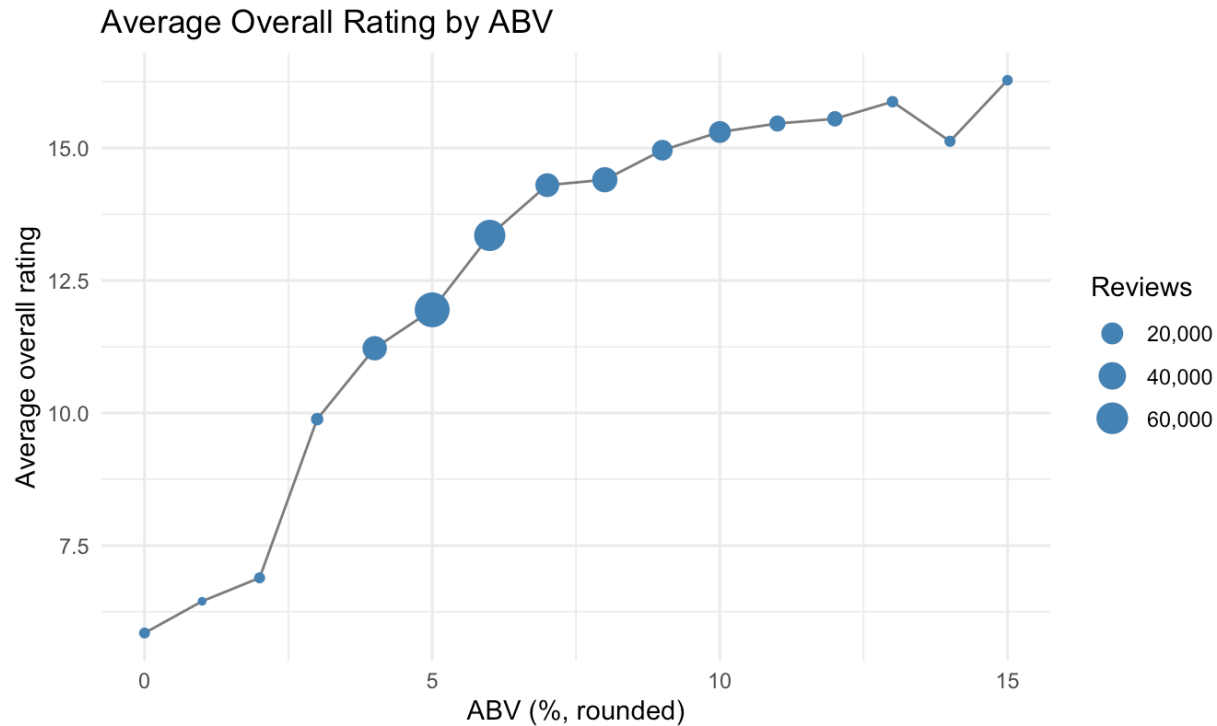
Even with transparency. What is the problem, and what did you ask for instead?

Second attempt: average by ABV



Correlation: 0.40. Describe the relationship. Does stronger **cause** better?

Rating vs. alcohol: reading it



- Ratings **rise** from about 11/20 at 3–4% to 15/20 at 9–10%, then **flatten**. Past 13% the points bounce: few beers live there (small dots)
- Overplotting fix from Tuesday: 275,000 points became 16 averages
- Causal? A 12% imperial stout is brewed with more care and reviewed by people who **sought it out**. “Stronger beers are rated higher” is a fact. “Alcohol makes beer better” is a story

What a strong answer looks like

The question (Task 6): *Describe the relationship. Does it flatten? Does higher alcohol cause higher ratings?*

Weak

“There is a positive relationship between ABV and rating. Correlation is not causation.”

Both sentences are true. Neither says anything about *this* data.

Strong

“Ratings climb from **11/20** at 4% to **15/20** at 10% ($r = 0.40$), then flatten; above 13% there are too few beers to say. I would not call this causal: strong beers are mostly imperial stouts and quadrupels, made with more ingredients and bought on purpose by enthusiasts, so style and reviewer both change with ABV.”

The recipe again: **the pattern → the number → the reason → what it means for the decision.**

Part 2: Tasks 7–9

What you do now

Task 7: the brewery table

One row per **brewery**: number of styles, number of reviews, average rating. Keep breweries with 10+ reviews. Styles vs. rating, with a fitted line.

Watch for: the unit of observation just changed. Where did the 289,000 reviews go?

Task 8: over time

Reviews per year and average rating per year.

Watch for: the data ends on **January 13, 2012**.

Task 9: the recommendation

Three bullets for a new craft brewery: which styles, how many. Each bullet rests on **one number** from a chart. One bullet must state a **limitation** of this data.

Hint: who writes reviews on RateBeer, and what does the data **not** contain (prices, volumes, costs)?

Optional extra chart: the share of reviews going to IPAs by year.

Predict first, then run, then explain one line back. Write the prediction **before** the chart; a wrong prediction is fine and gets full credit.

Give your assistant a memory

Every tool reads one file at the start of each chat: `CLAUDE.md` (Claude Code), `AGENTS.md` (Codex), `.github/copilot-instructions.md` (Copilot), `.cursor/rules` (Cursor). Standing instructions go there **once**.

The handout's Setup step 4 has the prompt. It tells the assistant, among other things:

- I am a beginner: comment each step, do not run code, I run it
- Print every chart in the chunk, never only `ggsave()`
- **After every request, append one line to `ai-log.md`**

That last rule makes the AI log write itself. No more “remember before you close the chat”. Two minutes now, and it works for every assignment this semester (copy the file into each new project folder).

Copilot follows the file only in agent mode. Copilot chat users keep the end-of-session log prompt.

Before you submit: the checklist

Open your knitted [ratebeer-report.html](#) in a browser and check:

1. **Name and 10-digit USC ID** under the title
2. **Every chart is visible inside the HTML.** Code but no chart means the chunk only saved a file. Ask the assistant to print the chart in the chunk. The grader receives one file, not your figures folder
3. Under every task: **prediction**, **answer** with a number, **explain it back** in your own words
4. [ai-log.md](#) written by the assistant **before** you closed the chat

Submit both files on Brightspace by **Sunday, Sept. 20, 11:59 pm.**

Handout: [w3-ratebeer-case.html](#). Thursday: regression.

Questions?